Sharp finite-sum bounds for the Riemann heat flow

Certified strip transfer, a finite window, and an infinite tail

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Abstract

We study a normalized sharp lower bound for the paired finite sums in Polymath's effective approximation to the Riemann heat flow. At heat time $157/1000$, we certify monotonic transfer in the imaginary coordinate throughout $[\sqrt{87677/2500000}, \sqrt{343/500}]$ for four Euler-mollified index ranges covering $690988 \le N \le 3840000$. The proof combines active-branch right-derivative inequalities, exact analytic cases and outward interval arithmetic over the actual integer parameters. A generic signed-sum example shows why the old plain-weight transfer cannot simply be inherited. One certified finite floor then gives a complete strip margin exceeding 0.0017752269101384707 in its natural cutoff window. A separate eleven-term mollifier, with enlarged convolution head, gives an infinite-tail margin exceeding 0.02303064 from $N = 3840000$ onward. An activation lemma bounds the loss of the sharp floor from one index of a leg to a larger one by a one-dimensional sum in which the Euler majorant cancels. It reduces the remaining finite floors to certified rows and outward activation bounds on a finite mesh, which a floating plan puts at twenty rows. Only the first row is certified and no activation bound has been evaluated outwardly, so the proposed bound $\Lambda \le 0.1745354$ does not follow from these results.

Draft status. This draft assembles the results and execution records of round-one Notes 01–02 [3, 4]. Published approximation results and the historical initial-verification and barrier inputs remain explicit imports. The computational checks of the other sections are internal, LLM-assisted checks by the drafting model. A cross-model LLM review of round one [5] has been written; its reconciliation and specialist human review remain pending. Section 7 was added after that review by the reviewing model: its proofs have had same-model checks only and await cross-model review, and its table is floating planning. The authoring configuration and assistance are recorded in the acknowledgements.

1 Purpose and normalization

The Riemann heat flow in the normalization of Polymath [1] is

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \tag{1.1}$$

where

$$\Phi(u) = \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}.$$

The present calculation concerns a quantitative finite-sum estimate at a fixed positive time. Its main result is a strip-transfer theorem for a sharp coefficient mass, not a criterion equivalent to

the Riemann hypothesis. The underlying sharp triangle inequality is inherited from the Polymath method and the preceding audit [1, Section 8.5][7, 6]; we give the native normalization and its proof because the application requires the paired coefficients and their changing absolute-value branches.

Fix throughout

$$t_* = \frac{157}{1000}, \quad y_0 = \sqrt{\frac{87677}{2500000}}, \quad y_1 = \sqrt{\frac{343}{500}}, \quad N_0 = 690988, \quad N_* = 3840000. \tag{1.2}$$

In particular $y_1^2 = 1 - 2t_*$. The finite Euler prime sets are specified by Table 1. All statements over a leg quantify over *every integer* in its indicated range.

Leg	Integer indices N	Euler prime set P
0	$690988 \le N \le 728999$	$\{2, 3, 5, 7, 11\}$
1	$729000 \le N \le 818999$	$\{2, 3, 5, 7\}$
2	$819000 \le N \le 1027999$	$\{2, 3, 5\}$
3	$1028000 \le N \le 3840000$	$\{2, 3\}$

Table 1: The finite legs. The last natural index also lies in the tail.

1.1 The imported effective approximation

For clarity we specify the functions used in the approximation. On $\mathbb{C} \setminus (-\infty, 1]$ use the principal logarithm and put

$$\mathcal{M}_0(s) = \frac{s(s-1)\sqrt{2\pi}}{16}\pi^{-s/2}\exp\left[\left(\frac{s}{2} - \frac{1}{2}\right)\operatorname{Log}\frac{s}{2} - \frac{s}{2}\right], \tag{1.3}$$

$$\alpha(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2}\operatorname{Log}\frac{s}{2\pi}, \qquad \mathcal{M}_t(s) = e^{t\alpha(s)^2/4}\mathcal{M}_0(s). \tag{1.4}$$

For $z = x + iy$, set

$$s_+ = \frac{1 + y - ix}{2}, \qquad s_- = \frac{1 - y + ix}{2}, \qquad B_t(z) = \mathcal{M}_t(s_+),$$

$$s_* = s_+ + \frac{t}{2}\alpha(s_+), \qquad \gamma = \frac{\mathcal{M}_t(s_-)}{\mathcal{M}_t(s_+)}, \tag{1.5}$$

$$\kappa = \frac{t}{2}\left[\alpha(s_-) - \alpha(\overline{s_+})\right], \qquad b_t(m) = e^{(t/4)\log^2 m}. \tag{1.6}$$

The natural cutoff and finite approximation are

$$N = \left\lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \right\rfloor, \tag{1.7}$$

$$f_t(x + iy) = \sum_{m=1}^{N} b_t(m) m^{-s_*} + \gamma \sum_{m=1}^{N} m^y b_t(m) m^{-\overline{s_*} - \kappa}. \tag{1.8}$$

Theorem 1.3 of [1], for $0 < t \le 1/2$, $0 \le y \le 1$ and $x \ge 200$, supplies

$$|H_t/B_t - f_t| \le e_A + e_B + e_{C,0}. \tag{1.9}$$

Its bounds for $|\gamma|$, $\operatorname{Re} s_*$ and $|\kappa|$ are the inputs used below. The carrier B_t is nonzero: s_+ is off the real cut, all factors in (1.3) are nonzero there, and the heat factor is an exponential. We retain the full $e_{C,0}$ allowance. The error supplier uses the conservative constant $10.50 = 3.58 + 6.92$ and denominator $x - 12$, as derived from Proposition 6.6(vi) in [1] and audited in [8]. No residual from a different corrected center is inserted.

1.2 Frozen windows and coefficients

At $t = t_*$ define

$$q_N = N^2 - \frac{t_*}{16}, \qquad x_N = 4\pi q_N, \qquad W_N = [x_N, x_{N+1}). \tag{1.10}$$

For $D = \prod_{p \in P} p$ and $d \mid D$, let

$$\lambda_d = (-1)^{\omega(d)} \prod_{p \mid d} b_{t_*}(p), \qquad \begin{aligned} B_{N,n} &= \sum_{\substack{d \mid D,\ d \mid n \\ n/d \le N}} \lambda_d b_{t_*}(n/d), \\ A_{N,n}(y) &= \sum_{\substack{d \mid D,\ d \mid n \\ n/d \le N}} \lambda_d b_{t_*}(n/d)(n/d)^y. \end{aligned} \tag{1.11}$$

Both leading coefficients are one. The frozen scalar bounds are

$$g_N(y) = \exp\Big(\frac{y}{50} - \frac{y}{2}\log q_N\Big), \tag{1.12}$$

$$\sigma_N(y) = \frac{1+y}{2} + \frac{t_*}{4}\log q_N - \frac{t_*}{2x_N^2}\left(1 - 3y + \frac{4y(1+y)}{x_N^2}\right)_+, \tag{1.13}$$

$$k_N(y) = \frac{t_* y}{2(x_N - 6)}, \qquad M_N(y) = \sum_{d \mid D} |\lambda_d| d^{-\sigma_N(y)}, \tag{1.14}$$

$$C_N(y) = \sum_{m=2}^{N} b_{t_*}(m)(m^{k_N(y)} - 1)m^{y - \sigma_N(y)}, \qquad K_N(y) = g_N(y)C_N(y). \tag{1.15}$$

The imported bounds imply throughout W_N

$$|\gamma| \le g_N < 1, \qquad \operatorname{Re} s_* \ge \sigma_N, \qquad |\kappa| \le k_N. \tag{1.16}$$

Indeed the first and third majorants decrease with x. In the lower bound for $\operatorname{Re} s_*$, the logarithm increases, and the product of x^{-2} with the nonnegative decreasing positive part decreases. This also handles its kink. These directions require no monotonicity of the final lower bound in independent scalar arguments.

2 The sharp native binding

For real B, A and $0 \le g < 1$, define

$$U_g(B, A) = \max\left(\frac{|B - gA|}{1 - g}, \frac{|B + gA|}{1 + g}\right) = \frac{|B - g^2 A| + g|A - B|}{1 - g^2}, \tag{2.1}$$

$$W_g(B, A) = (1 - g)U_g(B, A). \tag{2.2}$$

The normalized coefficient mass and lower bound are

$$V_N(y) = \sum_{n=2}^{DN} U_{g_N(y)}(B_{N,n}, A_{N,n}(y)) n^{-\sigma_N(y)}, \tag{2.3}$$

$$L_N^{\mathrm{sh}}(y) = \frac{(1-g_N(y))(1-V_N(y))}{M_N(y)} - K_N(y). \tag{2.4}$$

Lemma 2.1 (Paired real coefficients)**.** *Let $r = |\Gamma| < 1$. The real-linear map $T_\Gamma(w) = w + \Gamma\bar{w}$ satisfies $|T_\Gamma(w)| \geq (1-r)|w|$, and*

$$T_\Gamma^{-1}(Bz + \Gamma A\bar{z}) = \frac{(B - r^2A)z + \Gamma(A-B)\bar{z}}{1-r^2}.$$

In particular the modulus on the right is at most $U_r(B,A)|z|$. For fixed real A, B, $U_r(B,A)$ is nondecreasing in r.

Proof. The inverse formula follows by substitution. The triangle inequality and (2.1) give the modulus estimate. For monotonicity, simultaneous sign reversal reduces to $B \geq 0$. For $0 < r < 1$ the three formulas for U_r are

$$\frac{B - rA}{1-r}, \quad \frac{B + rA}{1+r}, \quad \frac{rA - B}{1-r}$$

on $A \leq B$, $B \leq A \leq B/r^2$, and $A \geq B/r^2$, respectively. Their derivatives are $(B-A)/(1-r)^2$, $(A-B)/(1+r)^2$ and $(A-B)/(1-r)^2$, all nonnegative on the corresponding branches. The formulas agree at their boundaries and extend continuously to $r = 0$. □

Proposition 2.2 (Binding in the approximation's native units)**.** *For an index and prime set in Table 1 and $y \in [y_0, y_1]$, if $L_N^{\mathrm{sh}}(y) > 0$, then*

$$|f_{t_*}(x+iy)| \geq L_N^{\mathrm{sh}}(y) \qquad (x \in W_N).$$

There is one division by M_N; the correction K_N is not divided again.

Proof. Write $f = F_B + \gamma F_{A,\kappa}$ for the two sums in (1.8), and remove $m^{-\kappa}$ to define $F_{A,0}$. For $E(s_*) = \prod_{p \in P}(1 - b_{t_*}(p)p^{-s_*})$, exact convolution gives

$$EF_B = \sum_{n=1}^{DN} B_{N,n} n^{-s_*}, \qquad \overline{E}F_{A,0} = \sum_{n=1}^{DN} A_{N,n} n^{-\overline{s_*}}.$$

Here $E \neq 0$, since $\sigma_N > 1.6 > (t_*/4)\log p$ makes each Euler-factor perturbation have modulus less than one. Put $\Gamma = \gamma E/\overline{E}$, so $|\Gamma| = |\gamma| = r$. Lemma 2.1, applied termwise with leading term $T_\Gamma(1)$, yields

$$|E(F_B + \gamma F_{A,0})| \geq (1-r)\left(1 - \sum_{n=2}^{DN} U_r(B_{N,n}, A_{N,n}) n^{-\operatorname{Re} s_*}\right).$$

If $V_N < 1$, (1.16) and monotonicity of U_r imply that this is at least $(1-g_N)(1-V_N)$. Also

$$|F_{A,\kappa} - F_{A,0}| \leq \sum_{m=2}^{N} b_{t_*}(m)(m^{k_N} - 1)m^{y-\sigma_N} = C_N.$$

Use $|E| \leq M_N$ and subtract $g_N C_N$ after division. Finally, $L_N^{\mathrm{sh}} > 0$ implies $V_N < 1$, since $M_N > 0$ and $K_N \geq 0$. □

Remark 2.3. The weight $W_g = \max(|B - gA|, (1-g)|B + gA|/(1+g))$ need not increase with g. The valid frozen-g argument uses U_g and positivity of $1 - V_N$. The improvement retains the paired real coefficients; a separate absolute bound on the two sums would recover the plain triangle loss.

3 Right derivatives and the transfer obligation

For real z, w set

$$\mathcal{D}(z,w) = \begin{cases} \operatorname{sgn}(z)w, & z \neq 0, \\ |w|, & z = 0. \end{cases}$$

It is the right derivative of the absolute value of a differentiable real function with value z and derivative w. We write D^+ for the upper right Dini derivative. Fix N, n, abbreviate $L = \log n$, and set

$$\ell = \frac{g_N'}{g_N} = \frac{1}{50} - \frac{1}{2}\log q_N < 0, \quad z_e = B_{N,n} + eg_N A_{N,n}, \quad d_e = 1 + eg_N, \quad v' = g_N(\ell A_{N,n} + A_{N,n}'). \tag{3.1}$$

The branch $u_e = |z_e|/d_e$ is active if it realizes the maximum in (2.1). Both branches are retained at a tie.

Lemma 3.1 (Active-branch gate)**.** *On the stated strip, $\sigma_{N,+}' \geq 1/2$, and*

$$D^+(u_e n^{-\sigma_N}) \leq \frac{n^{-\sigma_N}}{d_e}\left[\mathcal{D}(z_e, ev') - \left(\frac{L}{2} + \frac{eg_N'}{d_e}\right)|z_e|\right]. \tag{3.2}$$

Nonpositivity of the bracket for every active branch implies that the corresponding summand of V_N is nonincreasing. This remains valid at numerator zeros, branch ties and the positive-part kink of σ_N.

Proof. The argument of the positive part in (1.13) has negative y-derivative $-3 + 4(1+2y)/x_N^2$. Hence its contribution to the right slope of σ_N is nonnegative. Differentiate $|z_e|/d_e$ using $z_e' = ev'$ and $d_e' = eg_N'$, then use $\sigma_{N,+}' \geq 1/2$. The right derivative of a maximum of two locally Lipschitz branches is the maximum of their derivatives at the active branches. The resulting upper-right nonpositive derivative implies monotonicity; continuity includes the upper endpoint of the strip. □

Proposition 3.2 (The plain gate does not imply the sharp gate)**.** *In the generic real signed finite-exponential-sum class, a sufficient plain-weight Dini test does not imply the normalized sharp test.*

Proof. Take $n = 16$, $\sigma = \sigma_0 + y/2$, $g = 4^{-y}$, $B = 1$, and $A(y) = -1 + 3 \cdot 4^y - 16^y$. Thus $A(0) = B$. At $y = 1/2$, with $a = \log 2$,

$$g = \tfrac{1}{2}, \quad A = 1, \quad A' = -4a, \quad g' = -a.$$

The sufficient plain bracket is

$$g|A'| + g'|A| - \tfrac{1}{2}\log n\,(|B| + g|A|) = -2a < 0.$$

Both normalized branches equal one, but their weighted right slopes, after division by $16^{-\sigma}a$, are 2 and $-10/3$. The maximum is positive. The example refutes only generic automatic inheritance; it is not an example with the theta Euler coefficients of (1.11). □

4 The uniform finite-leg transfer certificate

Computer-assisted theorem 4.1 (Four-leg strip transfer)**.** *For every leg in Table 1, every integer N in that leg, and every $2 \leq n \leq DN$, the function*

$$y \longmapsto U_{g_N(y)}(B_{N,n}, A_{N,n}(y))n^{-\sigma_N(y)}$$

is nonincreasing on $[y_0, y_1]$. Consequently V_N is nonincreasing.

4.1 Divisor patterns and feasible integer coverage

Put $G = \gcd(n, D)$, and order its divisors as $1 = d_0 < \cdots < d_{r-1} = G$. The sector $n \le N$ has all divisors active. In sector $d_{j-1}N < n \le d_jN$, precisely $d_j, \dots, G$ are active. Writing $n = Gk$ imposes $\gcd(k, D/G) = 1$. If $n > GN$ there is no active divisor, and the summand is identically zero. The numbers of nonempty sector types are $3^{|P|}$, namely $243, 81, 27, 9$ on the four legs.

The certificate uses coordinates $h = \log N$, $v = \log(n/N)$, $L = h + v$. Define

$$\delta(h) = -\frac{1}{2}\log\left(1 - \frac{t_*}{16}e^{-2h}\right), \quad \ell = \frac{1}{50} - h + \delta(h), \quad F = g_N e^{yL} = e^{y(v+1/50+\delta(h))}. \tag{4.1}$$

Thus the full N-dependence is retained. For the active divisors let

$$r_d(L) = \exp\left[\frac{t_*}{4}\left(\sum_{p|d}\log^2 p + \log^2 d - 2L\log d\right)\right], \tag{4.2}$$

$$c = \sum_{d \text{ active}} (-1)^{\omega(d)} r_d(L), \qquad C_y = \sum_{d \text{ active}} (-1)^{\omega(d)} r_d(L) d^{-y}. \tag{4.3}$$

Then $B_{N,n} = b_{t_*}(n)c$ and $A_{N,n} = b_{t_*}(n)e^{yL}C_y$. Remove the positive factor $b_{t_*}(n)$ and put

$$Q = FC_y, \qquad T = F[(v + 1/50 + \delta)C_y + \partial_y C_y]. \tag{4.4}$$

The exact branch comparison is

$$u_- \ge u_+ \quad \Longleftrightarrow \quad (gc - Q)(c - gQ) \ge 0. \tag{4.5}$$

With $g = g_N$, the implemented branch predicate is nonpositivity of

$$(1 + eg)\mathcal{D}(c + eQ, eT) - \left[\frac{L}{2}(1 + eg) + e\ell g\right]|c + eQ| \tag{4.6}$$

for every possibly active branch. Multiplication by $1 + eg > 0$ preserves the sign in (3.2).

The interval cover embeds each actual integer pair in closed boxes for (h, v, y); sector closures only enlarge the domain. Outward exponentiation bounds the possible integers N, n. The exact sector inequalities and gcd restriction provide necessary feasibility tests. If at most 33 possible multiples $n = Gk$ remain, the code checks all of them against the integer leg/sector constraints and outer ratio bounds. A cell is discarded only if these necessary conditions admit no pair. A unique possible n tightens the interval for L to its logarithm. Closed bisection children share their boundary. These operations cannot remove a feasible pair or leave a subdivision gap.

4.2 An equivalent predicate at absolute-value kinks

Put

$$R = c - gQ, \quad S = Q - gc, \quad R' = -g(\ell Q + T), \quad S' = T - \ell g c.$$

The identity

$$\max(u_-, u_+) = \frac{|R| + |S|}{1 - g^2}$$

gives the equivalent derivative numerator

$$\mathcal{D}(R, R') + \mathcal{D}(S, S') - \left[\frac{L}{2} - \frac{2g^2\ell}{1 - g^2}\right](|R| + |S|). \tag{4.7}$$

A nonpositive enclosure of (4.7) proves the gate even when separate branch enclosures overestimate a tie. In either form, a zero-containing numerator retains the upper bound on its absolute right derivative. In (4.6) the coefficient of the plus absolute value can be negative; its complete signed interval product is retained. No midpoint sign is used to suppress a branch.

4.3 Analytic cases

Lemma 4.2 (A single active divisor)**.** *Every feasible sector with exactly one active divisor satisfies the gate.*

Proof. For that divisor d, put $m = n/d$, $a = \log m$. Then $B \neq 0$, $A = Be^{ay}$ and $0 \leq a \leq \log N < -2\ell$. The final inequality is $q_N/N > e^{0.04}$, which holds at the smallest N and persists because $N - t_*/(16N)$ increases. If $a > 0$ only plus is active and

$$\frac{d}{dy}\log u_+ = \frac{a}{2} + \frac{1}{2}\left[(\ell + a)\tanh\frac{y(\ell + a)}{2} - \ell\tanh\frac{y\ell}{2}\right] \leq \frac{a}{2}.$$

Indeed $x\tanh(yx/2)$ is even and increases with $|x|$, and $|\ell + a| \leq |\ell|$. Subtracting $\sigma'_{N,+}\log n$ bounds the weighted rate by $-\frac{1}{2}\log d \leq 0$. At $m = 1$, both branches equal $|B|$ identically and their weighted rates are also nonpositive. □

Lemma 4.3 (The isolated prime zeros)**.** *If a full-active $G = p \in P$ cell has $n = p$ as its only possible integer, the gate holds on every actual pair in that cell.*

Proof. Here $B = 0$, $A = b_{t_*}(p)(p^y - 1) > 0$ and minus is uniquely active. The weighted logarithmic rate is at most

$$\frac{\ell}{1-g} + \frac{\log p}{1 - p^{-y}} - \frac{\log p}{2} \leq \ell + \frac{\log p}{2} + \frac{1}{y} < 0.52 - 13 + 1.5 + \frac{1000}{187} = -\frac{52663}{9350} < 0.$$

Use $q_N > N^2/2$, $\log N > 13$, $\log p < 3$ and $y > 0.187$; the inequality $1/(1 - e^{-u}) \leq 1 + 1/u$ gives the middle estimate. No other simultaneous coefficient zero is assigned this shortcut. □

4.4 Execution and proof of the theorem

The engine uses 40-digit directed Decimal arithmetic. The nearest-mode correctly rounded exponential, logarithm and square root are padded outward by 10^{-38}. A cell passes by (4.6), (4.7), Lemma 4.2 or Lemma 4.3; it may also be proved empty. Otherwise it is subdivided or reported unresolved. Table 2 gives the complete pattern partition.

Leg	Pattern indices	Certificate	Remaining cells
0	$0, \ldots, 171$	First pass, complete prefix	None in prefix
0	$172, \ldots, 242$	Completion, full original roots	None
1	$0, \ldots, 80$	First pass	None
2	$0, \ldots, 26$	First pass	None
3	$0, \ldots, 8$	First pass	None

Table 2: Composition of the finite-leg transfer certificate.

Proof of Theorem 4.1. The root boxes and necessary integer tests cover every nonempty actual term. The first run completed legs 1–3 and the indicated prefix of leg 0 before reaching its 30,000-box cap. Pattern 172 was only partly evaluated then, so that work is not used for composition. The second run restarted its full root and completed the entire suffix 172–242. Its wrapper checks the engine hash, the first record's completed prefix and the other complete legs, and records the first record's hash. The exact prefix/suffix union is all 243 patterns, with no unresolved leaf in the completed domains. Each run used at most 45 seconds and 30,000 evaluations per leg, at depth at most 36; the measured combined interval time was 91.06 seconds and peak memory was below 22 MiB. The records in Appendix A summarize the interval decisions and contain accepted-leaf digests; the engine regenerates the leaf inequalities. No conclusion is drawn from a positive upper interval bound or a partial pattern. The sign predicates and the analytic cases therefore prove every active gate. Apply Lemma 3.1, and add the finitely many nonincreasing summands. □

The first execution record alone has a false full-completion flag, correctly reflecting its budget stop. Both records are necessary for Theorem 4.1. A separate same-model check compares actual integer terms with 110-digit direct arithmetic, tests that feasible toy pairs survive pruning, and rejects the tied adverse example of Proposition 3.2. These checks examine implementation; they do not replace the uniform interval cover or establish review independence. No global extremal slack is inferred from the accepted leaf thresholds or their count.

Corollary 4.4 (Transfer of a positive native floor)**.** *If $V_N(y_0) < 1$, then $L_N^{\mathrm{sh}}(y) \ge L_N^{\mathrm{sh}}(y_0)$ for $y_0 \le y \le y_1$.*

Proof. Theorem 4.1 makes V_N nonincreasing. The factor $1 - g_N$ is positive and increasing, and M_N is nonincreasing because $\sigma'_{N,+} \ge 1/2$. Finally put $\rho_N = t_*/[2(x_N - 6)]$. Each positive summand of K_N has logarithmic rate at most

$$\ell + \tfrac{1}{2}\log N + \rho_N \log N + 1/y_0.$$

This follows from $\frac{d}{dy}\log(m^{\rho_N y} - 1) \le \rho_N \log m + 1/y$. For $t_* < 0.162$, $N \ge 690988$,

$$\ell < 0.52 - \log N, \qquad \rho_N < 0.081/N^2,$$

so the rate is less than $0.52 - 6.5 + 0.081/N + 1000/187 < -0.631$. Thus K_N decreases, and every operation in (2.4) has the required direction when $1 - V_N(y_0) > 0$. □

5 A finite window and the complete approximation allowance

The approximation allowance is recomputed at the new exact time; the historical time box is not reused. A convenient statement is

Proposition 5.1 (Uniform approximation budget)**.** *For every natural cutoff $N \ge N_0$, $x \in W_N$ and $y_0 \le y \le y_1$ at $t = t_*$, the inherited effective approximation supplies*

$$e_A + e_B < 2.159 \cdot 10^{-12}, \qquad e_{C,0} < 2.829422 \cdot 10^{-7},$$
$$e_A + e_B + e_{C,0} < 2.829444 \cdot 10^{-7} < 3 \cdot 10^{-7}.$$

Proof. We instantiate the audited general U1–U5 reduction [8, Section 4] at $t = t_*$, not its old fixed-parameter conclusion. Here is the budget explicitly. Put $z_0 = t_*/(16N_0^2)$, $L_0 = \log q_{N_0}$, $x_0 = x_{N_0}$ and

$$\delta_0 = \frac{t_*}{4}\frac{z_0}{1 - z_0} + \frac{t_*}{2x_0^2}, \qquad \sigma_0 = \frac{1 + y_0}{2} + \frac{t_*}{2}\log N_0 - \delta_0.$$

Use the endpoint cap of Lemma 6.1 below, with $m_0 = 2000$, to form

$$S_0 = 1 + \sum_{n=2}^{m_0} b_{t_*}(n) n^{-\sigma_0} + \mathrm{Cap}_{t_*}(m_0, N_0; \sigma_0),$$
$$P_0 = 1 + e^{1/50}(1 - z_0)^{-1/2} \exp\left[\frac{t_* \log N_0}{2(x_0 - 6)}\right],$$
$$U_0 = \frac{(t_*^2/16)L_0^2 + 0.626}{x_0 - 6.66}.$$

The error supplier bounds $e_A + e_B$ by $P_0 S_0 (e^{U_0} - 1)$. The C-term allowance is bounded by

$$\exp\left[-\frac{1+y_0}{4}L_0 - \frac{t_*}{16}L_0^2 + \frac{1.24(3^{y_1} + 3^{-y_1})}{N_0 - 1/8} + \frac{3\sqrt{L_0^2 + \pi^2/4} + 10.50}{x_0 - 12}\right]. \tag{5.1}$$

The head/cap monotonicity and domain gates of U1–U5 are rechecked in `TAIL_T157_M600000_RECORD.json`; its two U3 sign gates are less than -0.42558 and -0.47805. Directed evaluation gives the respective upper bounds $2.158316897 \cdot 10^{-12}$ and $2.829421823 \cdot 10^{-7}$. The reduction is uniform for all $N \geq N_0$ and the full new strip. The error formulas are imports from [1, 8]; the parameter substitution and enclosure are recorded in [3] and Appendix A. □

Computer-assisted theorem 5.2 (First-window strip margin)**.** *At $t = t_*$, throughout $x \in W_{690988}$ and $y_0 \leq y \leq y_1$,*

$$\left|\frac{H_{t_*}(x+iy)}{B_{t_*}(x+iy)}\right| > 0.0017752269101384707. \tag{5.2}$$

Proof. The outward fixed-point pilot encloses

$$0.0017755269101384707588 < L^{\mathrm{sh}}_{690988}(y_0) < 0.0017755269101384708095.$$

It uses the exact enumeration $n = Gk$, $\gcd(k, D/G) = 1$, signed convolutions and the weight W_g. In the record, S denotes the *unnormalized* mass $(1-g)V_N$, and $Q = 1 - g - S$; this S must not be mistaken for V_N. The 110-bit fixed-point sum allowance is below $4.54 \cdot 10^{-20}$, and the scalar/table supplier uses inherited 45-digit directed arithmetic. Error propagation uses the 1-Lipschitz absolute-value and maximum maps, including the multiplier $(1-g)/(1+g)$ and directly enclosed $G^{-\sigma_N}$ scalars. The record's positive numerator implies $V_N(y_0) < 1$. Corollary 4.4 and Proposition 2.2 transfer this floor through the entire strip and window. Subtract the full $3 \cdot 10^{-7}$ allowance of Proposition 5.1 using (1.9). □

6 An infinite-tail estimate with a larger head

This section re-instantiates the *generic arguments* of the preceding tail audit [8, Sections 3.1–3.8]; its old stated time interval does not contain t_*. The partition and reduction are given here, and every new application gate is reevaluated.

Lemma 6.1 (Endpoint cap)**.** *For $t \geq 0$, integers $1 \leq a < N$ and $\sigma > (t/2)\log N$,*

$$\sum_{a < n \leq N} b_t(n) n^{-\sigma} \leq \mathrm{Cap}_t(a, N; \sigma) := \log(N/a) \max_{u \in \{a, N\}} e^{(1-\sigma)\log u + (t/4)\log^2 u}.$$

Proof. The summand decreases, so its sum is bounded by the integral from a to N. With $u = e^v$, the integrand is $\exp((1-\sigma)v + tv^2/4)$, a convex exponent whose maximum on the logarithmic interval is at an endpoint. This is the subrange version of Polymath's Lemma 8.2 [1]. □

The tail uses a different rational mollifier. Put $\mathcal{S} = \{1,2,3,4,5,6,7,10,11,13,14\}$ and, in that order, set

$$(\nu_d)_{d\in\mathcal{S}} = \left(1, -\frac{1021}{1000}, -\frac{1054}{1000}, -\frac{9}{200}, -\frac{1119}{1000}, \frac{1001}{1000}, -\frac{1043}{1000}, \frac{128}{125}, -\frac{161}{200}, -\frac{447}{500}, \frac{456373}{500000}\right). \tag{6.1}$$

These coefficients are not the finite Euler coefficients λ_d. Let $\mathcal{E}_\nu(s) = \sum_{d\in\mathcal{S}} \nu_d d^{-s}$ and fix $M = 600000 < N_*$. With $c_m = \sum_{d\mid m,\, d\in\mathcal{S}} \nu_d b_{t_*}(m/d)$, the exact pair partition is

$$\begin{aligned}\mathcal{E}_\nu(s) \sum_{k\le N} b_{t_*}(k)k^{-s} = 1 + \sum_{m=2}^{M} c_m m^{-s} \\ + \sum_{d\in\mathcal{S}} \nu_d d^{-s} \sum_{\lfloor M/d\rfloor < k\le N} b_{t_*}(k)k^{-s}.\end{aligned} \tag{6.2}$$

All pairs with $dk \le M$ occur in the head, since $M < N$; the rest occur in exactly one routed sum. The historical nonnegative term `OV` bounds part of this routed remainder a second time and is not needed in this partition.

Let $x_* = x_{N_*}$ in this section only, and put

$$z_* = \frac{t_*}{16N_*^2}, \qquad \widehat{\delta} = \frac{t_*}{4}\frac{z_*}{1-z_*} + \frac{t_*}{2x_*^2}, \qquad \widehat{k} = \frac{t_*}{2(x_*-6)}, \tag{6.3}$$

$$\sigma_1(N,y) = \frac{1+y}{2} + \frac{t_*}{2}\log N - \widehat{\delta}, \qquad \sigma_2(N,y) = \sigma_1(N,y) - y - \widehat{k}, \tag{6.4}$$

$$G(N,y) = \exp\Big(\frac{y}{50} - \frac{y}{2}\log(N^2 - t_*/16)\Big). \tag{6.5}$$

The imported approximation bounds imply $\operatorname{Re} s_* \ge \sigma_1$, $|\kappa| \le \widehat{k}$ and $|\gamma| \le G$ for every $N \ge N_*$. Here $-\log(1-z) \le z/(1-z)$ and the positive part in (1.13) is bounded by one. Define

$$P = \sum_{m=2}^{M} |c_m| m^{-\sigma_1}, \qquad \mathfrak{M} = \sum_{d\in\mathcal{S}} |\nu_d| d^{-\sigma_1}, \tag{6.6}$$

$$T_R = \sum_{d\in\mathcal{S}} |\nu_d| d^{-\sigma_1} \operatorname{Cap}_{t_*}(\lfloor M/d\rfloor, N; \sigma_1), \tag{6.7}$$

$$A_B = G\left[\sum_{n=1}^{M} b_{t_*}(n) n^{-\sigma_2} + \operatorname{Cap}_{t_*}(M, N; \sigma_2)\right], \tag{6.8}$$

$$D' = P + T_R + \mathfrak{M} A_B. \tag{6.9}$$

The partition and ordinary triangle inequality give

$$|\mathcal{E}_\nu(s_*) f_{t_*} - 1| \le D', \qquad |f_{t_*}| \ge \frac{1-D'}{\mathfrak{M}} \quad (D' < 1). \tag{6.10}$$

The second inequality follows because $|\mathcal{E}_\nu| \le \mathfrak{M}$; $D' < 1$ itself ensures that the mollifier and the product are nonzero.

Proposition 6.2 (Order of the tail reductions)**.** *The majorant $D'(N,y)$ is nonincreasing first in y, then in N at the lower y endpoint, provided the recorded cap gates below hold. The contraction's A-cap all-index gate is needed only near y_0; the error caps still require the full strip hull.*

Proof. For fixed N, all B-side terms decrease with y through $\partial_y\sigma_1 = 1/2$. Every G-weighted A-side head term or cap endpoint $u \le N$ has logarithmic y-rate at most

$$Y_M(N) = \frac{1}{50} - \frac{1}{2}\log(N^2 - t_*/16) + \frac{1}{2}\log N < 0.$$

The logarithmic cap width is independent of y. Thus D' decreases throughout the full strip, before any N reduction.

For a fixed cap left endpoint a, set $q = \log N$, $r = \log a$, and $\sigma(N) = \beta + (t_*/2)q$. The two cap candidates have logarithmic q-derivatives

$$\frac{1}{q-r} - \frac{t_*}{2}r, \qquad \frac{1}{q-r} + 1 - \sigma(N).$$

Both are negative if

$$\frac{t_*}{2}r(q-r) > 1, \qquad (\sigma(N)-1)(q-r) > 1. \tag{6.11}$$

The left sides increase with q once they hold. At the reduced lower y value these gates therefore make all routed and A caps decrease with N. The heads and mollifier majorant decrease as their exponents increase; G also decreases. This proves the contraction reduction. The approximation-error supplier does not use this y reduction, so its two error caps must retain the full-hull all-index gates. □

Computer-assisted theorem 6.3 (Tail at the unchanged starting index)**.** *For $t = t_*$, every $N \ge N_*$, $x \in W_N$ and $y_0 \le y \le y_1$,*

$$|H_{t_*}(x+iy)/B_{t_*}(x+iy)| > 0.02303064. \tag{6.12}$$

Proof. The contraction is enclosed at $N = N_*$ on

$$I_{\mathrm{box}} = [1872719/10^7, 23409/125000],$$

containing y_0. The interval's lower endpoint lies below y_0, so Proposition 6.2 reduces the required strip to that box. The error supplier retains

$$I_{\mathrm{ext}} = [1872719/10^7, 8282512/10^7],$$

which covers y_1; the historical top 0.8231039 would not suffice. Directed evaluation at the exact new time gives Table 3. The least routed cap gates exceed 1.93875 and 1.45491; the A-cap second gate on I_{box} exceeds 1.10728. Its historical extended-hull test does not pass (its computed lower endpoint is about 0.51236); that test is not needed after the stated reduction order. Both full-hull error caps pass, with least gate exceeding 1.97475. The validity gates $\sigma_j > (t_*/2)\log N$ and all remaining domain and monotonicity gates are retained in the record.

The exact head is $M = 600000$; the two error heads use 3000 terms. Their inherited error formulas, including the 10.50 weld, give the complete tail allowance in Table 3. Equations (1.9) and (6.10), and the all-index reductions, then prove the result. All numerical comparisons are reevaluations of the generic audited arguments at t_*, not applications of the old fixed-time theorem outside its domain. □

Quantity at the declared hull	Safe outward bound
P	< 0.214842556622
T_R	< 0.214622245029
$\mathfrak{M}$	< 1.631076048739
A_B	< 0.326760019216
D'	< 0.962435242680
$e_A + e_B + e_{C,0}$ in the tail	$< 1.4900275280 \cdot 10^{-8}$
$(1 - D')/\mathfrak{M} - (e_A + e_B + e_{C,0})$	> 0.02303064473

Table 3: New-time tail budget with $M = 600000$. The final row uses the full-precision enclosures, rather than recombination of rounded table entries.

At the unchanged head 153814, the same new-time budget has an upper bound about 1.15923637 for D' and does not close. This is a failure of that head and majorant to certify contraction, not a theorem of nonvanishing failure or an obstruction to other heads. Similarly, retaining the redundant `OV` padding in the successful case would prevent this particular budget from closing. The raw historical checks and their replacement reasons are preserved; adverse required gates are not removed.

7 Activation along a leg

Corollary 4.4 transfers the floor of one index through the strip, but no result above relates the floors of different indices. This section bounds the change from one side. Passing from an index N_a to a larger index N of the same leg moves the frozen scalars in the favorable direction and activates the divisor pairs (d, m) with $N_a < m \le N$. The first effect is discarded; the second costs a one-dimensional sum in which the Euler majorant cancels.

Fix a leg of Table 1, with prime set P and $D = \prod_{p \in P} p$, a point $y \in [y_0, y_1]$, and integers $N_a \le N$ of the leg. Write g_a, σ_a, k_a and M_a for the quantities (1.12)–(1.14) at the index N_a and the point y, and put

$$\Delta_y(N_a, N) = \sum_{N_a < m \le N} b_{t_*}(m)\, m^{-\sigma_a} \Big[W_{g_a}\big(1, m^y\big) + g_a \big(m^{k_a} - 1\big) m^y \Big], \tag{7.1}$$

with $W_g = (1 - g)U_g$ as in (2.2). The summands are nonnegative, so $\Delta_y(N_a, N)$ is nondecreasing in N and vanishes at $N = N_a$.

Lemma 7.1 (Activation bound)**.** *If $L^{\mathrm{sh}}_{N_a}(y) - \Delta_y(N_a, N) > 0$, then*

$$L^{\mathrm{sh}}_N(y) \ge L^{\mathrm{sh}}_{N_a}(y) - \Delta_y(N_a, N).$$

Proof. Directions in N. As N increases, q_N and x_N increase. Since $y > 0$, this gives $g_N \le g_a$ and $k_N \le k_a$. In (1.13) the logarithm increases, and the subtracted product of x_N^{-2} with the nonnegative, nonincreasing positive part decreases. Hence $\sigma_N \ge \sigma_a$, and $M_N \le M_a$ because every $d \mid D$ satisfies $d \ge 1$. These are the directions behind (1.16), read in N rather than in x.

Frozen scalars. Let

$$V_{\mathrm{hyb}} = \sum_{n=2}^{DN} U_{g_a}(B_{N,n}, A_{N,n}(y))\, n^{-\sigma_a}, \qquad C_{\mathrm{hyb}} = \sum_{m=2}^{N} b_{t_*}(m)(m^{k_a} - 1) m^{y - \sigma_a}$$

be V_N and C_N formed with the coefficients of N and the scalars of N_a, and put

$$L_{\mathrm{hyb}} = \frac{(1 - g_a)(1 - V_{\mathrm{hyb}})}{M_a} - g_a C_{\mathrm{hyb}}.$$

The monotonicity in Lemma 2.1 and the directions above give $V_N \le V_{\mathrm{hyb}}$ and $0 \le g_N C_N \le g_a C_{\mathrm{hyb}}$. If $V_{\mathrm{hyb}} \le 1$, then also $0 \le 1 - V_{\mathrm{hyb}} \le 1 - V_N$, $0 < 1 - g_a \le 1 - g_N$ and $0 < M_N \le M_a$, so that $L_N^{\mathrm{sh}}(y) \ge L_{\mathrm{hyb}}$.

Activated pairs. A pair (n, d) with $d \mid D$ and $d \mid n$ contributes to the coefficients (1.11) of N but not to those of N_a exactly when $m = n/d$ satisfies $N_a < m \le N$. Hence

$$\big(B_{N,n}, A_{N,n}\big) = \big(B_{N_a,n}, A_{N_a,n}\big) + \sum_{\substack{d \mid D,\ d \mid n \\ N_a < n/d \le N}} \lambda_d \, b_{t_*}(n/d)\big(1, (n/d)^y\big),$$

where $B_{N_a,n} = A_{N_a,n} = 0$ for $n > DN_a$. As the larger of two weighted absolute values of linear forms, U_g is a seminorm on $\mathbb{R}^2$. Therefore

$$U_{g_a}(B_{N,n}, A_{N,n}) \le U_{g_a}(B_{N_a,n}, A_{N_a,n}) + \sum_{\substack{d \mid D,\ d \mid n \\ N_a < n/d \le N}} |\lambda_d| \, b_{t_*}(n/d) \, U_{g_a}\big(1, (n/d)^y\big).$$

Summation. Multiply by $n^{-\sigma_a}$ and sum over $2 \le n \le DN$. The first terms give V_{N_a}. The pairs (n, d) in the second correspond, through $n = dm$, to the pairs (d, m) with $d \mid D$ and $N_a < m \le N$, and $n^{-\sigma_a} = d^{-\sigma_a} m^{-\sigma_a}$. Since $\sum_{d \mid D} |\lambda_d| d^{-\sigma_a} = M_a$,

$$V_{\mathrm{hyb}} \le V_{N_a} + M_a \sum_{N_a < m \le N} b_{t_*}(m) \, U_{g_a}(1, m^y) \, m^{-\sigma_a}.$$

Also $C_{\mathrm{hyb}} = C_{N_a} + \sum_{N_a < m \le N} b_{t_*}(m)(m^{k_a} - 1) m^{y - \sigma_a}$. Inserting both relations in L_{hyb}, the factor M_a cancels and $L_{\mathrm{hyb}} \ge L_{N_a}^{\mathrm{sh}}(y) - \Delta_y(N_a, N)$.

Conclusion. If the right side is positive, so is L_{hyb}. As $g_a C_{\mathrm{hyb}} \ge 0$, this forces $V_{\mathrm{hyb}} < 1$, and the second step gives the claim. □

The proof uses only Lemma 2.1, the directions of (1.16) in N and the seminorm property of U_g. The same argument applies to the plain weight $|B| + g|A|$, also a seminorm, with $W_g(1, u)$ replaced by $1 + gu$. Frozen quantities over runs of indices also occur in the historical producer [7, Section 3]; the lemma is an explicit bound of that kind for the sharp weight. Only $y = y_0$ is needed below, because Corollary 4.4 transfers the floor of every index through the strip.

Lemma 7.2 (Activation sums at y_0)**.** *Let $N_a \le N$ lie in one leg, take g_a, σ_a, k_a at $y = y_0$, and put*

$$\theta_a = \frac{1 - g_a}{1 + g_a}, \qquad \eta = k_a \log N \, N^{k_a}, \qquad S(s) = \sum_{N_a < m \le N} b_{t_*}(m) \, m^{-s}.$$

Then

$$\Delta_{y_0}(N_a, N) \le \theta_a \, S(\sigma_a) + g_a(\theta_a + \eta) \, S(\sigma_a - y_0). \tag{7.2}$$

For all integers $N_a = N^{(0)} < N^{(1)} < \cdots < N^{(J)} = N$ and $s \in \{\sigma_a, \sigma_a - y_0\}$,

$$S(s) \le \sum_{j=1}^{J} \mathrm{Cap}_{t_*}\big(N^{(j-1)}, N^{(j)}; s\big). \tag{7.3}$$

Proof. For $N_a < m \le N$, $g_a^2 m^{y_0} = e^{y_0/25}(m/q_{N_a})^{y_0}$. Within each leg $m/q_{N_a} < 3.7 \cdot 10^{-6}$, the extreme case being the two ends of leg 3, and $0.187 < y_0 < 0.1873$. Hence

$$g_a^2 m^{y_0} < e^{0.0075}(3.7 \cdot 10^{-6})^{0.187} < 0.1.$$

For $u \geq 1$ and $g^2 u \leq 1$, (2.1) gives $U_g(1,u) = (1 - g^2 u + g(u-1))/(1-g^2) = (1+gu)/(1+g)$, so $W_{g_a}(1, m^{y_0}) = \theta_a(1 + g_a m^{y_0})$. Moreover $m^{k_a} - 1 \leq N^{k_a} - 1 \leq \eta$, since $e^v - 1 \leq v e^v$ for $v \geq 0$. Inserting these in (7.1) gives (7.2).

For (7.3), apply Lemma 6.1 on each subrange $(N^{(j-1)}, N^{(j)}]$. Its hypothesis $s > (t_*/2)\log N^{(j)}$ holds for both values of s, because $N^{(j)} \leq N$, $s \geq \sigma_a - y_0$, and $\sigma_a - y_0 > (t_*/2)\log N$. For the last inequality, the positive part in (1.13) and the difference $(t_*/4)\log q_{N_a} - (t_*/2)\log N_a$ change σ_a by less than 10^{-13}. As $N/N_a < 3.74$ within a leg,

$$\sigma_a - y_0 - \frac{t_*}{2}\log N > \frac{1-y_0}{2} - \frac{t_*}{2}\log\frac{N}{N_a} - 10^{-13} > 0.406 - 0.104 - 10^{-13} > 0. \qquad \square$$

Corollary 7.3 (Mesh criterion)**.** *Let a leg have first index N_{first} and last index N_{last} in Table 1, and let $N_{\text{first}} = N_1 < N_2 < \cdots < N_r \leq N_{\text{last}}$. Put $E_i = N_{i+1} - 1$ for $i < r$ and $E_r = N_{\text{last}}$. Suppose that for each i there are numbers $\ell_i \leq L^{\text{sh}}_{N_i}(y_0)$ and $\delta_i \geq \Delta_{y_0}(N_i, E_i)$ with $\ell_i - \delta_i > 3 \cdot 10^{-7}$. Then $L^{\text{sh}}_N(y_0) > 3 \cdot 10^{-7}$ for every integer N of the leg.*

Proof. Every N of the leg lies in exactly one interval $[N_i, E_i]$. There $\Delta_{y_0}(N_i, N) \leq \Delta_{y_0}(N_i, E_i) \leq \delta_i$, so $L^{\text{sh}}_{N_i}(y_0) - \Delta_{y_0}(N_i, N) \geq \ell_i - \delta_i > 3 \cdot 10^{-7}$, and Lemma 7.1 applies. $\square$

Once its hypotheses are certified on all four legs, Corollary 7.3 supplies the premise of Corollary 8.1. A certificate consists of certified lower ends ℓ_i, from the pilot of Theorem 5.2 run at each mesh index, and upper bounds δ_i from (7.2) and (7.3) in directed arithmetic, which needs only the frozen scalars of each mesh index. The enclosure of Theorem 5.2 is the first row of leg 0.

Remark 7.4 (A planning mesh, in floating point only). Table 4 lists a mesh found in float64 by a greedy rule: from the first index N_i of a leg, take the largest E for which the bound δ of (7.2) on $(N_i, E]$, with a single cap $J = 1$ in (7.3), satisfies $L^{\text{sh}}_{N_i}(y_0) - \delta > 3 \cdot 10^{-7}$, and restart at $E + 1$. Bounded by a single cap, the last span of leg 3 would need a seventh row, at $N = 3550349$; a partition into 16 geometric pieces closes it instead (eight pieces suffice, four do not). The floating direct sum over that span is 0.148783. Twenty rows result, the largest at $N = 1880119$. With floating direct sums in place of every cap bound, the counts per leg are again 4, 5, 5 and 6. Within a leg the floors grow by factors between 1.77 and 2.05 from one row to the next, and the first span of leg 0 has 3683 indices.

These are planning values, not a certificate. Only the first row of Table 4 is certified, and no bound δ_i has been evaluated outwardly. On interior spans the greedy rule leaves small margins in the last column, down to $1.1 \cdot 10^{-10}$. They still exceed the difference between floating and certified floors, which is below 10^{-16} at $N = 690988$ and $N = 1974864$; a certificate may nevertheless prefer spans with slack. Two floating checks of the chain in the proof of Lemma 7.1 give

(N_a, N)	$L^{\text{sh}}_N(y_0)$	L_{hyb}	$L^{\text{sh}}_{N_a}(y_0) - \Delta_{y_0}(N_a, N)$
$(690988, 691988)$	0.002152	0.001539	0.001294
$(1028000, 1030000)$	0.005306	0.004498	0.004273

In both cases Δ_{y_0} is about twice the loss $L^{\text{sh}}_{N_a}(y_0) - L_{\text{hyb}}$ at frozen scalars, and the movement of the scalars, which the lemma discards, more than offsets that loss. As a cost measurement, the unchanged pilot certified

$$0.163564430553564191107 < L^{\text{sh}}_{1974864}(y_0) < 0.163564430553564191115$$

at the largest row of the plan with direct sums, in 247 seconds with a peak of 339 MiB, inside its 600-second and 512 MiB guards. Every row of Table 4 lies below that index, and the pilot's work and memory grow with N. The files behind these values are listed in Appendix A.

Leg	N_i	E_i	$L^{\mathrm{sh}}_{N_i}(y_0)$	δ_i	$L^{\mathrm{sh}}_{N_i}(y_0) - \delta_i - 3 \cdot 10^{-7}$
0	690988	694670	0.00177553	0.00177494	$2.9 \cdot 10^{-7}$
	694671	701277	0.00315769	0.00315719	$2.0 \cdot 10^{-7}$
	701278	713204	0.00561378	0.00561307	$4.1 \cdot 10^{-7}$
	713205	728999	0.00997173	0.00723317	$2.7 \cdot 10^{-3}$
1	729000	734947	0.00262914	0.00262846	$3.8 \cdot 10^{-7}$
	734948	745859	0.00475959	0.00475925	$3.6 \cdot 10^{-8}$
	745860	766077	0.00861023	0.00860983	$1.0 \cdot 10^{-7}$
	766078	804221	0.0155531	0.0155527	$5.7 \cdot 10^{-8}$
	804222	818999	0.0280188	0.00556659	$2.2 \cdot 10^{-2}$
2	819000	828799	0.00358388	0.00358325	$3.3 \cdot 10^{-7}$
	828800	847621	0.00675051	0.00674996	$2.5 \cdot 10^{-7}$
	847622	884344	0.0126958	0.0126953	$2.2 \cdot 10^{-7}$
	884345	958159	0.0238072	0.0238067	$2.7 \cdot 10^{-7}$
	958160	1027999	0.0443923	0.0197452	$2.5 \cdot 10^{-2}$
3	1028000	1046970	0.00477606	0.00477567	$8.3 \cdot 10^{-8}$
	1046971	1086871	0.00974623	0.00974578	$1.4 \cdot 10^{-7}$
	1086872	1173175	0.0198154	0.0198149	$2.1 \cdot 10^{-7}$
	1173176	1370858	0.0399830	0.0399826	$1.0 \cdot 10^{-7}$
	1370859	1880118	0.0794341	0.0794338	$1.1 \cdot 10^{-10}$
	1880119	3840000	0.152894	$0.150563^{\dagger}$	$2.3 \cdot 10^{-3}$

Table 4: Floating planning mesh for Corollary 7.3; not a certificate (Remark 7.4). The floors and the bounds δ_i are float64 values, and the last column is computed before rounding. Each δ_i is the bound (7.2) with a single cap in (7.3), except †, with 16 geometric pieces. Leg 0 begins at the certified row of Theorem 5.2.

8 The remaining finite bridge and the conditional target

Polymath's Theorem 1.2 [1] converts three inputs into $\Lambda \le t_0 + y_0^2/2$: initial verification to height $X/2$, final-time nonvanishing for $x \ge X + \sqrt{1-y_0^2}$ and $y_0 \le y \le \sqrt{1-2t_0}$, and an intermediate curved barrier. For

$$X = 6000000185827,$$

the Platt–Trudgian verified height [2] covers $X/2$. The new curved barrier at $t_0 = t_*$ is contained in the historical closed box

$$[X, X+1] \times [0.1809, 1] \times [0, 0.16125].$$

Indeed its horizontal width is less than one, its lower ordinate is at least $y_0 > 0.1809$, and $t_* < 0.16125$. The historical barrier is inherited on the status recorded by its audit [9]; it is not regenerated here. The first natural window still contains $X + \sqrt{1-y_0^2}$, and the half-open W_N tile without gaps.

Corollary 8.1 (Explicit remaining obligation)**.** *Assume the imported initial-verification and historical barrier inputs just described. If a valid new floor certificate establishes*

$$L^{\mathrm{sh}}_N(y_0) > 3 \cdot 10^{-7} \quad \textit{for every integer } 690989 \le N < 3840000$$

with its leg's coefficients and all producer or amortization errors paid, then $\Lambda \le 0.1745354$.

Proof. The first floor is Theorem 5.2; the remaining finite floors transfer by Corollary 4.4 and pay Proposition 5.1. Theorem 6.3 supplies the infinite exterior. These cover the entire final-time region. Apply the imported criterion and compute $t_* + y_0^2/2 = 872677/5000000 = 0.1745354$. □

The premise about the remaining floors is still open. Corollary 7.3 reduces it to certified rows and outward activation bounds on a finite mesh of each leg, and the floating plan of Table 4 uses twenty rows; only the first of them is certified. Neither the historical plain-weight rows at time 0.16125, the favorable sampled sharp rows, the planning values of Table 4, nor the transfer theorem proves the premise. No monotonicity in N is assumed; Lemma 7.1 bounds the change in N from one side. This draft therefore reports a completed transfer calculation, one finite-window margin, a revised infinite exterior and a reduction of the finite bridge to a finite certificate, not a complete certificate of the conditional conclusion.

The control-family screen is also quantitative. The generic paired algebra has no shape premise that excludes colliding sources. Euler-factor controls change the coefficients and conductor; weighted-square, reflected, positive-primitive and earlier positive-source controls do not provide the complete theta approximation with its stated error allowance [3]. Their own representations and errors would have to replace these inputs. A valid method may certify a collision-free box of a control, but cannot produce a positive complete margin on a region containing its collision. The present theorem makes no such claim.

A Evidence and reproduction

All current evidence below is under `papers/investigations-3/02_sharp_finite_sum_bounds/numerics/round-1/` in the research checkout. The source notes are cited as [3, 4]. Files and imported helper pins are bound by the build record accompanying this manuscript.

File	Role
`certify_sharp_pilot.py`	Outward single-row implementation; one native floor.
`SHARP_PILOT_RECORD_20261017.json`	First floor and fixed-point/scalar allowances.
`check_sharp_y_transfer_gate.py`	Actual-integer domain cover and both derivative predicates.
`SHARP_Y_TRANSFER_GATE_RECORD_20261017.json`	Three complete legs and the largest-leg prefix.
`complete_sharp_y_transfer_gate.py`	Checks the complete prefix and starts the original suffix roots.
`SHARP_Y_TRANSFER_COMPLETION_RECORD_20261017.json`	Completed suffix and exact union; required with the preceding record.
`check_gate_arithmetic.py`	Separate same-model implementation checks; not uniform coverage.
`check_t157_tail_budget.py`	Hash-pinned generic error/tail adapter with explicit substitutions.
`TAIL_T157_M600000_RECORD.json`	Uniform error allowance and successful new-time tail.
`TAIL_T157_M153814_RECORD.json`	Inherited-head budget diagnostic.

Table 5: The small evidence package. Floating scout values are diagnostics only.

The computations behind Section 7 are in the subfolder `claude-review/`:

- `review_checks_claude_20261017.py` and its record `REVIEW_CHECKS_CLAUDE_20261017.json`, whose section F gives the floating rows, meshes and hybrid-row checks of Remark 7.4; `REPLAY_ON_EDWARDS_MACHINE_20261017.json` compares a replay on a second host, identical in every compared field;
- `activation_plan_claude_20261018.py` and its record `ACTIVATION_PLAN_CLAUDE_20261018.json`: the side conditions of Lemma 7.2 and the bounds δ_i of Table 4;

- `COST_PROBE_PILOT_N1974864_CLAUDE_20261017.json`: the cost measurement of Remark 7.4.

These are planning and measurement files, not certificates.

The transfer engine checks its interval primitives, exact rational one-sided derivative identities, and toy integer membership. The additional checker compares 1080 actual term points at 110 digits, tests 171 feasible toy pairs through 2223 containing child cells, and rejects the adverse tied control. These counts account for checks; the proof uses analytic arguments and the interval signs, not the number of checks. The tail supplier uses 50-digit directed arithmetic with 10^{-48} transcendental padding, and explicitly retains failed legacy comparisons whose historical target, floor or reduction has changed.

From the numerical directory `papers/investigations-3/02_sharp_finite_sum_bounds/numerics/round-1/`, a cheap implementation check is

```
python3 check_gate_arithmetic.py --output /tmp/sharp-gate-check.json
```

From its subfolder `claude-review/`, the planning of Section 7 is reproduced by

```
python3 activation_plan_claude_20261018.py --output /tmp/activation-plan.json
```

The exact bounded-run commands and plans are in the numerical README. The completion wrapper uses the preserved first record beside the engine; it does not automatically consume an arbitrary freshly generated first pass. Resource stops remain incomplete and must be reported. Each source file and record is below 1 MiB; no large row corpus is created. Hashes establish input and execution identity, while arithmetic and the stated analytic reductions establish the inequalities.

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Section 7, its table and the corresponding changes to the abstract, the draft status, Section 8, Appendix A and the bibliography were drafted for Edward Baker with LLM assistance from Anthropic's Claude Opus 5.5 (configured identifier `claude-opus-5-5`) in Claude Cowork, reasoning effort `max`, following the round-one cross-model review [5]. A same-model subagent checked the proof of Lemma 7.1 for that review, and another checked Section 7 as drafted, its proofs and Table 4 against the records. Neither check is independent review, and the section awaits cross-model review.

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[5] Prepared for E. Baker with LLM assistance, *Round-one cross-model review of the sharp finite-sum bounds*, 17 October 2026, Claude Opus 5.5 (`claude-opus-5-5`), effort `max`. Investigation 02, review 01; it cites this manuscript at commit `efba736`, before Section 7 was added, `../../reviews/01_ROUND_ONE_CROSS_MODEL_REVIEW_CLAUDE_20261017.md`.

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[8] Internal LLM audit, *Tail lemma and uniform error budget: audit of the remaining analytic suppliers of hypothesis (ii)*, 12 October 2026. `papers/investigations-2/newman_certificate_audit/notes/4_TAIL_LEMMA_AND_UNIFORM_ERROR_BUDGET_AUDIT_20261012.md`.

[9] Internal LLM audit, *Barrier certificate audit*, 13 October 2026. `papers/investigations-2/newman_certificate_audit/notes/5_BARRIER_CERTIFICATE_AUDIT_20261013.md`.